

# SAMUEL AMOS-OSEBEYO

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## Education

### Imperial College London

Expected Graduation: May 2028

*Electronic and Information Engineering: MEng (integrated Master's)*

## Research and Work Experience

### Adaptive and Intelligent Robotics Lab

June 2026 – Sep 2026

*Undergraduate Researcher, Legged Locomotion*

- Trained quadruped parkour policies in NVIDIA Isaac Lab with massively parallel GPU-accelerated PPO (rsl\_rl), under EurekaVerse-style LLM-generated terrain curricula.
- Ported the lab's damage-adaptive locomotion pipeline from Isaac Gym to Isaac Lab and rewrote the damage catalogue and training scheme, completing 20% more goals under simulated actuator and joint faults than the previous implementation.
- Deployed the damage-adaptive policy to a Unitree Go2, where it performs parkour from a single depth camera, and measured in simulation the loss introduced by the privileged-scandot-to-depth distillation step.
- Implementing World Model-based Perception, a recurrent state-space model over depth and proprioception, so damage-robust behaviour is learned directly from deployed sensing rather than distilled, collapsing the fault catalogue into one generalist policy.

### Dexory

June 2025 – Sep 2025

*Junior Electronics Intern*

- Designed, brought up and validated multi-layer PCBs and modern C++ firmware for embedded control and sensor subsystems on autonomous warehouse robots.
- Built modular C++ firmware libraries coordinating I2C communication and telemetry collection, increasing sensor data throughput by 30%.
- Executed system-level testing of electromechanical assemblies, characterising power draw and reducing bus communication latency.

### Imperial Planetary Robotics Lab

Oct 2024 – May 2025

*Junior Robotics Engineer*

- Led the hardware redesign and ROS2 integration of a 6-DOF robotic arm for a Mars-rover platform, developing embedded C++ firmware, multi-layer PCBs and a MoveIt inverse-kinematics configuration; the team finished 16th globally and 1st in the UK out of 100+ teams.

## Projects

### Sample-Efficient RL for Goal-Conditioned Manipulation (PyTorch, panda-gym)

- Implemented Soft Actor-Critic with Hindsight Experience Replay from scratch and compared four arms: SAC+HER 52% at 2.3M environment steps, behaviour cloning 95%, BC-then-RL fine-tuning 100% at 1.6M, half the steps of demonstration-seeded SAC+HER at equal out-of-distribution generalisation.
- Diagnosed the baseline's ceiling by instrumenting per-episode object height and established that naively loading behaviour-cloned weights into SAC destroys a 93% policy within eight gradient updates, fixed with a BC-anchored actor loss scaled by the critic's value scale.

### Franka Panda VR Teleoperation and Data Collection (Hackathon)

- Engineered a low-latency VR teleoperation pipeline driving a physical Franka Panda arm from a Meta Quest 3 via ADB/Oculus Reader.
- Architected a distributed controller with ZeroRPC and Polymetis (real-time control over libfranka), running directly on Franka hardware, and collected over 100 demonstration trajectories in LeRobot format for VLA and HIL-SERL policy training.

### Autonomous Self-Balancing Registration Robot (Team Lead, six-person group project)

- Led six engineers building a two-wheeled inverted-pendulum robot that explores an unknown venue, seeks out people and identifies them by face, and integrated all five subsystems against a fixed demo deadline.
- Co-developed the ROS 2 localisation and mapping stack, fusing wheel-encoder odometry and gyroscopic yaw with LiDAR scan matching in an EKF, and replaced explore\_lite with a custom frontier explorer after measuring 2s map-update latency and repeated loss of localisation.
- Owned the perception pipeline: an SCRFD detector feeding 512-dimensional ArcFace embeddings matched by cosine similarity, reaching over 80% recognition accuracy at 1140ms detection-to-embedding latency.

## Skills

**Languages and Tools:** Python | C++ | C | PyTorch | Isaac Lab | Isaac Gym | rsl\_rl | ROS2 | OpenCV | Linux | Git

**Learning and Control:** Reinforcement learning (PPO, SAC, HER) | Imitation learning | World models | Sim-to-real

**Perception and Hardware:** SLAM | EKF sensor fusion | Multi-layer PCB design | Embedded C++ firmware